

Defining the racial and ethnic “other”: Constructing an American identity through visualizing census data in the U.S. *Statistical Atlases*

Journal of Technical Writing and

Communication

2022, Vol. 52(4) 460–492

© The Author(s) 2022

Article reuse guidelines:

sagepub.com/journals-permissions

DOI: 10.1177/00472816221086523

journals.sagepub.com/home/jtw

Li Li 

Abstract

This study analyzes the visualization of census data in the U.S. *Statistical Atlases* from 1874 to 1925. I examine how visual strategies were used to construct an American identity by contrasting the “native” population with the “other”—new immigrants and African Americans, which were visualized as undesirable counterparts. By defining the “other,” the Atlases created a pan ethnic identity of the “native white” population, established a racial hierarchy, and hardened the division between old and new immigrants. The study develops a rhetorical framework for understanding how data design is used to marginalize racially and ethnically minority groups.

Keywords

data visualization, visual rhetoric, statistical atlas, census data, historical study, race, ethnicity, american identity, design, discrimination

Data visualization makes raw data accessible to wider audiences. In today’s age of “big data,” professional communication researchers have emphasized the significant role of data visualization in meaning making and in influencing audience’s beliefs and attitudes (Frith, 2017; Graham et al., 2015; Jones, 2015; Kostelnick, 2016; Rawlins and Wilson, 2014; Wolfe, 2015; Zhang, 2017). A growing number of scholars have

Elon University, Elon, NC, USA

Corresponding Author:

Li Li Department of English, Elon University, 2338 Campus Box, Elon, NC 27244, USA.

Email: lli@elon.edu

explored data visualization as rhetorical artifact that embodies ideologies and values (see Barton and Barton, 1993/2004; Brasseur, 2005; Kimball, 2006; Kinross, 1989; Kostelnick, 2004; Rawlins and Wilson, 2014; Welhausen, 2015). In the recent years, the social justice turn in technical communication field has called for more scholarship in revealing power dynamics and acknowledging inequities such as racism, xenophobia, and sexism in communication practices (Agboka, 2013; Colton and Holmes, 2018; Williams and Pimentel, 2014). Williams and Pimentel (2014) ask us to “uncover those communicative practices used to negatively impact historically marginalized groups and identify new practices” (pp. 1–2). In response to this call for social justice, I present a visual analysis of the U.S. *Statistical Atlases* in the 19th and 20th centuries through the critical lens of race and ethnicity to demonstrate how data visuals are used to marginalize racial and ethnical minorities. The purpose of the study is to show how minority groups, especially African Americans and new immigrants who were not Northern and Western Europeans, were visualized in the *Atlases* of the late 19th century, and make visible discriminatory practices and racist language used in data visualization. I hope today’s professional communicators can be more aware of the inequities in communication practices and make more ethical decisions in creating their own data graphics.

In this article, I examine the rhetorical framing in one of the most well-known practices of data visualization in American history: *Statistical Atlases*, specifically the six *Atlases* in the late 19th and early 20th centuries. The U.S. *Statistical Atlases* are collections of data displays (i.e. charts, graphs, and maps) based on government census data of population, trade, and industries (Friendly, 2008). The US Census Bureau has a long tradition of visualizing and disseminating national census results since 1874. The Bureau published six consecutive *Statistical Atlases* from 1874 to 1924. As stated in the first *Atlas* of 1874, the *Atlases* were produced “for distribution to public libraries, learned societies, colleges and academies” (Walker, 1874, p. 1). Unfortunately, the publication of *Statistical Atlases* was stopped after the 1924 *Atlas* and did not resume until the 1970s. The focus of analysis in this study is the first six *Atlases*, whose graphic achievements are much acclaimed than the later ones. The six *Atlases* are “an exquisite sampler of all the graphical methods known at the time” with each *Atlas* adapting “to the problem at hand” (Friendly, 2008, p. 18), such as racial and immigration disputes, at its own particular historical moment.

As a powerful method to convey information about the nation, the *Statistical Atlases* do much more than simply reflect social reality; rather, they play a key role in the construction of that reality. Censuses, where the *Statistical Atlases* were based, have served as highly effective ways of turning knowledge into power (see Anderson & Fienberg, 1999; Hannah, 2000, 2001). The knowledge of the population produced by the census gave those in power insight into social conditions, allowing them to know the population, categorize them, set boundaries, and devise appropriate plans to manage the population. Although statistics are normally presented and perceived as scientific and objective, the entire process of census-taking, from designing

the survey instrument, to interpret the data, to disseminating the results, is infused with political ideologies (Balzhiser et al., 2019; Kertzer and Arel 2002; Urla 1993;).

As a part of the census process, the *Atlases* were employed by the government to construct the nation as we “imagined” it to be, as Benedict Anderson (2016) points out in his influential book *Imagined Communities*. Anderson identifies the census, the map, and the museum as three institutions of power used by the state to shape the common imagination of its dominion—“the nature of the human beings it ruled, the geography of its domain, and the legitimacy of its ancestry” (p. 164). Anderson also argues that print capitalism made the connection between people in a nation possible and “through print, formed in their secular, particular, visible invisibility, the embryo of the nationally imagined community” (p. 44). The publishing of the census data in the form of *Statistical Atlases* serves to build an imagined national community. The people in the national community were mapped out in the *Atlases*, hence were conceptualized as sharing, with a certain number of “others,” a common collective identity as Americans. The *Statistical Atlases*, in this respect, emerged as the most visible, and arguably the most politically important, means by which states statistically depict a collective identity.

Many scholars have recognized *Atlases* as significant graphical innovations in the 19th century and exemplary examples of data visualizations even today (see Friendly, 2008; Funkhouser, 1937; Kostelnick, 2004; Monmonier, 1994). However, little scholarly attention has been paid to the *Statistical Atlases* as an embodiment of ideologies and values. In his article “Melting-pot ideology, modernist aesthetics, and the emergence of graphical conventions: The *Statistical Atlases* of the United States, 1874–1925,” Charles Kostelnick studies the *Statistical Atlases* as the embodiment of aesthetic values and cultural conventions of the era. He argues that the *Atlases* are “artificial constructs that project a reading of the nation at a specific historical moment, and in that sense those constructs are highly rhetorical, even argumentative” (p. 227).

Informed by Kostelnick’s work, I study the *Atlases* as powerful rhetorical artifacts generated by the US government to construct a collective American identity in the 19th and 20th centuries. Inspired by Barton and Barton’s study on map design and ideology, I develop a rhetorical framework for understanding how visual language can be deployed to frame minority groups as marginal and undesirable. I argue that particular naming, classifying, and visualizing methods are used to identify the unfavorable “others” in the national population. Visual strategies such as color coding, spatial placement, grouping and scattering of visual objects are used in the *Atlases* for racialization and marginalization. A monolithic American identity is constructed in the *Statistical Atlases* by contrasting the so called “native” population—older immigrants (western and northern Europeans); with the “other”—newer immigrants (eastern and southern Europeans, Asian immigrants), and African Americans, who were visualized as undesirable counterparts. By charting the new immigrants and African Americans as “others” in the *Atlases*, the compilers created a pan ethnic identity for the “native White”¹ population, establishing a racial hierarchy of the White over the Blacks and

an ethnic hierarchy of Anglo-Saxon immigrants over eastern and southern Europeans and Asians.

More broadly, by confronting this history, this study informs technical communicators on various design strategies for visualizing big data, which may be used to advance political ideologies in today's world. *Statistical Atlases* were created in late 19th and early 20th centuries, a period of intensive immigration and racial disputes. Today, these disputes remain at the forefront of political controversies calling for scholarly investigation of how American identity is constructed rhetorically. While data design technologies have evolved dramatically since the 19th century, the graphical coding strategies of data, such as coloring and layout, remain essential to today's data designers. The rhetorical framework of data design developed in the study may inform further critical studies on contemporary data graphics. In the followings, I situate the study in the social justice scholarship in the technical communication field and frame the methodology of the study. Next, I discuss the historical context in which the *Atlases* were created, and then analyze the visual strategies in the *Atlases*.

Race and Ethnicity in Technical Communication

Scholars started a cultural critique of power dynamics in technical communication practices in the 1990s (Blyler, 1998; Herndl, 1993; Katz, 1992, 1993; Longo, 1998; Slack *et al.*, 1993). Longo (1998) argues that scholarship in the field should account for “struggles for knowledge legitimation ... influenced by institutional, political, economic, and/or social relationships, pressures, and tensions” (p. 61). In 2006, Scott and Longo's special issue of *Technical Communication Quarterly* extends the field's critical scholarship by calling for making the “cultural turn” to address the relationships between power/knowledge and ethics within professional communication. In the same year, Scott, Longo, and Wills (2006) published the collection *Critical Power Tools: Technical Communication and Cultural Studies*. The collection of essays aims to “historicize technical communication's role in hegemonic power relations—approaches that are openly critical of nonegalitarian, unethical practices and subject positions, that promote values other than conformity, efficiency, and effectiveness, and that account for technical communication's broader cultural conditions, circulation, and effects.” (p. 1).

Continuing the work of the “cultural turn” of the early 2000s, many scholars have called for a paradigm shift of technical communication to issues of social justice, diversity, equity, and inclusion (Agboka 2013; Colton and Holmes, 2018; Jones, 2016; Jones and Walton, 2016; Savage and Mattson, 2011). Jones (2016) called for scholars and practitioners alike to adopt a social justice framework. Jones and Walton (2016) defined such a framework as “social justice research in technical communication [which] investigates how communication broadly defined can amplify the agency of oppressed people—those who are materially, socially, politically, and/or economically under-resourced” (p. 347). Social justice research helps “communicators understand how to avoid sexism, racism, able-ism, age-ism, and other prejudices within research,

teaching, and professional practices.” (Colton and Holmes, 2018, p. 5). A growing number of scholars have addressed social justice and diversity issues in technical communication including African-Americans’ use of technology (Banks, 2006; William, 2010), the representation of Latinos in American technical communication (Johnson, Pimentel, and Pimentel, 2008; St. Germanie-Madison, 2006; Pimentel and Balzhiser, 2012; Balzhiser et al., 2019), critical research methods in international technical communication (Agboka 2013, 2014), and race and ethnicity in the technical communication curriculum design (Haas, 2012). Statements from Association of Teachers of Technical Writing (ATTW) and other professional organizations in light of 2020’s anti-Black and anti-Asian racism movements called for a redress of racism in our field (see Haas, 2020; Haas et al., 2021).

Despite the call for social justice in technical communication, only a few scholars have taken up this task concerning data visualization. Because of the seemingly neutrality of data visuals, the effectiveness and efficiency of data visuals has often been the focus of data visual researchers (Bertin, 2001; Cleveland and McGill, 1987; Tufte, 1990; Ware, 2013). However, more and more scholars started to problematize the effectiveness approach and critique data visuals as rhetorical artifacts that promote social, cultural, and political ideologies (Barton & Barton, 1993/2004; Brasseur, 2005; Frith, 2017; Kimball, 2006; Kinross, 1989; Kostelnick, 2004, 2017; Rawlins & Wilson, 2014; Welhausen, 2015; Zhang, 2017).

Barton and Barton’s important work “Ideology and the map: Toward a postmodern visual design practice” reveals the ideological mechanism of seemingly objective data visuals. They examine the design of maps and show how the rules of inclusion and exclusion create ideological dominance, marginalize and de-power the “others.” Kimball also claims that data visuals “inherently privilege the viewpoints of the powerful” (Kimball, 2006, p. 379) and he critiques the use of color coding in London poverty maps that constructed the problem of poverty as a fact with the historical context of the 19th century. Kostelnick and Hassett (2003) argue that data visuals embody social and cultural conventions. Social conventions regularize visual language to create hierarchies that affects readers’ interpretation of a visual (Kostelnick 2017). Welhausen (2015) studies early yellow fever maps in the social and cultural contexts of the 19th century and shows how visual conventions and scientific authority invoked power and authority. In an effort to resist such privileging and repressing, Hepworth and Church (2018) propose a data visualization ethics methodology that asks researchers to consider the ethics of their data workflow, selected visualization formats, and individual design choices.

Methodology

Much social justice research in technical communications tends to focus on how information visualization can entail inequities by establishing discriminatory hierarchical systems that impact its audience. Situated within this pool of scholarships, this study extends the existing scholarships to the exploration of how xenophobia and

racism function in data design. Barton and Barton's inspiring work on the ideological underpinnings of map design serves as the framework of my methodology. Barton and Barton identify that the "rules of inclusion" and "rules of exclusion" work in mapmaking to privilege certain interests and repress the others. The "rules of inclusion" determines "whether something is mapped, what aspects of a thing are mapped and what representational strategies and devices are used to map those aspects" (Barton and Barton, 1993/2004, p. 235). In terms of design strategies, privileging can be achieved through organizational arrangements such as placing items at the top or the center of a page, or by placing items at the top of an ordered list; while repressing can be achieved through excluding items, renaming them for diminishing purposes, and placing them in the margins. These rules explicitly or implicitly claim power and legitimate dominate interests. Barton and Barton's methodology on data graphics advocates for social justice and calls for change in visual communication. As Mirel comments, Barton and Barton's goal of visual rhetoric is to enhance "democratic practices, inclusionary meanings, and strategic consultative communications." (Mirel, 2004, P. 233)

Inspired by Barton and Barton's methodology, I analyze the *Statistical Atlases* from the perspectives of the "others," populations that are relegated to a repressed category and are often silenced and marginalized. I ask the following research questions:

1. How were the racial and ethnic "other", such as African Americans and newer immigrants from regions other than northern and western Europe, represented in the *Atlases*? What kind of American identity was constructed/imagined through the visualization of census data in the *Atlases*? How did the representation reflect and shape the historical and political context of the era when the *Atlases* were designed?
2. What design strategies are used to privilege the dominate interests while repress the "other's" interests in the data visuals? Is there a consistent framework of design strategies used in the *Atlases* that can inform today's technical communicators?

Building on Barton and Barton's rules of inclusion and exclusion, I examine what are included and excluded in the naming and classification of the American "natives" and "other" population in the data graphs. Expanding Barton and Barton's methods on map design, I study the graphic coding of data visuals in the *Atlases* including coloring, shading, spatial placements, grouping and separation of data. By synthesizing these design strategies, I develop a rhetorical framework for understanding how data designs privilege and repress interests in these historical *Atlases* but may also be meaningful for today's technical communicators.

Before the analysis of the visual language in the *Statistical Atlases*, I examine the creation of the *Atlases* in its social historical contexts and the dominant racial and ethnic discourses in the 19th century.

Historical Factors in Representing the “Other”

The Creation of the Statistical Atlases

The power of the *Atlases* to portray a nation depends on large part on the rise of social statistics and census-taking. The broad enthusiasm in the 19th century for statistics came from a growing belief in the power of facts (Mezey, 2002), and many believed that obtaining concrete data about the population could end political disagreements for the common good (Mezey, 2002). During the late nineteenth century, when social statistics grew dramatically and were popularized, the census came to play an important role in documenting and defining the form and content of the nation. At the same time, the biggest tidal wave of immigrants rushed into the United States in the mid 19th century. The government sensed the urgent need to examine and manage the fast-growing population. The census was then the most important tool used to illustrate the “virtues or vices of particular regions, peoples, or ways of life in America” (Mezey, 2002, p. 1711). Official state statistical offices were established to design, collect, analyze, and report census data in the United States and across Europe in the 1800s (Friendly, 2008, pp. 13–14). The census and the *Statistical Atlases* derived from the census started to play a crucial role in American political and social life.

The accumulation of large data through census-taking created the need for the invention of visual representations to manage the data sets. The 19th century is called the “golden age” of statistical graphics due to their explosive growth (see Friendly, 2008; Funkhouser, 1937). Most of the modern forms of data display were invented in the 19th century, such as histograms, time-series plots, and scatterplots (Friendly, 2008). The first national *Statistical Atlas* of the U.S. was created in 1874 under the supervision of Francis A. Walker, the superintendent of the Census Office in 1870 and 1880. Subsequently, five consecutive *Statistical Atlases* were published in the late 19th and early 20th centuries.

The *Statistical Atlases* translated, for the first time in American history, the complex census data into accessible and appealing visual displays for the general public. As Margo J. Anderson (1999) states, “the *Atlas* allowed the general public to see at a glance demographic characteristics that only the statistically trained could easily glean from the dry compilations of numbers in the tables themselves” (p. 92). The six *Atlases* generated new audiences (both political and popular) because of its appealing visual representation and accessibility to the general public. The *Atlases* pictured the changing nation literally rather than metaphorically (Mezey, 2002), providing a vivid picture of the country as imagined by the nation state.

The Racial and Ethnic Discourse in Which The Statistical Atlases Were Created

The *Atlases* were created during a racially and ethnically dynamic moment in American history. For most of the 19th century and before, immigration to the United States was free, open and unrestricted by the government (Daniels, 2002). However, gradually older immigrant groups in the nation felt threatened or even frightened by the arrival of

millions of newcomers. Nativism and prejudice against new immigrants, especially non-Whites, became a new sentiment of the nation in the late 19th century. The “old” immigrants, predominately from northern and western Europe, drew a line between themselves and the “new” immigrants, mostly from southern and eastern Europe and from Asia. In 1882, the Chinese Exclusion Act was passed to deny naturalization to Chinese in the United States and to refuse the entry of returning Chinese immigrants who temporarily left the country. The anti-immigration movement gained its final success with the Immigration Act of 1924. This bill directly controlled the number of incoming immigrants and remained in effect until mid-1960s. The bill operated on a national quota system based on the percentage of the nationality in the existing American population of the 1890 census, clearly attempting to maintain the ethnic composition of the time before a large number of non-Anglo-Saxon immigrants poured into the country. This Act assigned the largest quotas to Great Britain (about half of the total admissions), followed by Germany and Scandinavia. Italians had fewer than 6,000 seats, and Greeks, Slavs, and Jews got only a very small share (Purcell, 1995). Immigrants from Asia were excluded except for a very small number of Japanese and Filipinos.

On the other hand, the Civil War (1861–1865) and Emancipation Proclamation brought changes to the racial dynamics in the country. Formerly free Blacks and newly freed slaves were now entitled American citizens with voting rights and other privileges. However, White supremacists also strove to reamplify their views and oppose the recognition of Black political rights. At the century’s end, Blacks were largely disenfranchized, despite of the Fifteenth Amendment, and subject to pervasive public and private segregation, discrimination, and violence, despite of the Fourteenth Amendment and the Civil Rights Act of 1866 (Nobles, 2000).

The racist discourse prevalent in the 19th century argues that since Blacks were naturally inferior to Whites, they were as unfit to exercise their rights as citizens and were thereby disqualified from full participation in American economic, political, and social life (Nobles, 2000). The Black inferiority and White superiority ideology was sustained by racial science in the 19th century. The rise of data science, especially the use of census, largely informed and advanced racial science. Justifications for racism in the United States have relied heavily on racial census data. As the historian Thomas F. Gossett (1997) observes, “[A] striking feature of the literature of racism.... is the patient way in which the racists explain ‘scientific fact’ to their opponents” (p. 253). The census and *Statistical Atlases* became basic ways to educate the political and public audiences on racial science. Created by and embedded in institutional census processes, the *Statistical Atlases* directly contributed to the public racial discourse.

The Creator of the Statistical Atlases

The initiator of the *Statistical Atlases*, Francis A. Walker, was “one of the preeminent statisticians in the United States” (Hannah, 2000, p. 80). He was the Chief of the

Bureau of Statistics, President of the American Statistical Association, and Superintendent of Census 1870 and 1880. Beyond his remarkable work in statistics, Walker was also one of the principal experts on the economic and demographic policies of the country, especially on immigration (Schor and Schor, 2017). He was a fervent advocate of immigration restriction and published a series of articles and books on the racial competition between “new immigrants” and the “native Americans.”

In “Restriction of Immigration,” an article in the June *Atlantic Monthly* in 1896, Walker condemns the “worsening character of immigrants” and warns that newer immigrants who were not Anglo-Saxons, threatened “Americans’ peace and safety.” His most well-known theory, the “race suicide,” stated that immigration is a “replacement of native by foreign elements” (Walker, 2006, p. 84). He reasons: “They [“native Whites”] became increasingly unwilling to bring forth sons and daughters who should be obliged to compete in the market for labor and in the walks of life with those whom they did not recognize as of their own grade and condition” (p. 86). He argues for restricting immigration at large, since “American institutions, the American rate of wages, the American standard of living, are brought into serious peril” (p. 94). He exerted great influence on the advocates of immigration restriction and contributed to the passage of America’s first immigration restrictive policy: the Immigration Act of 1924.

Walker based his argument on statistics, especially the census data, which was a novelty in 19th century. He claimed, by examining census data, that the native-born birth rate was lower than that of the foreign-born leading to the gradual disappearance of Anglo-Saxon stock of Americans. Census data was used to establish the differences between the “new immigrants” and the “native Americans.” His reasoning was adopted by representatives of immigrant communities and by the federal government to bring about legislative change. It was not a coincidence that Walker and his followers used the language of “race” to describe the competition between the new immigrants and the “natives.” The racial construction of the new immigrants promoted the ideologies of xenophobia along with racism that prevailed in the 19th century.

In the following sections, I will examine the verbal and visual strategies used in the *Atlases* that reflected and shaped the racial discourse and promoted xenophobia in the late 19th and early 20th centuries.

Naming and Categorizing The “Others”

As crucial steps of census taking and data visualization, categorization and naming the population categories are two “othering” strategies in the *Atlases*. Race and ethnicity are not objective categories which censuses simply count, “but a fluid and internally contradicting discourse, partly created by and embedded in institutional processes, including those of the census itself” (Nobles, 2000, p. 2). The naming of population categories in the *Atlases* lends legitimacy to the official constructions of the

“native” identity of America. Through naming, minority groups are forced into existence or erasure.

The construction of a “native” identity started with the naming of the “others” inhabited in the country. The non-native others were defined by the Immigration Act of 1924, the first Act in history that restricted the incoming immigrants by their nationalities and races. The Act stipulated:

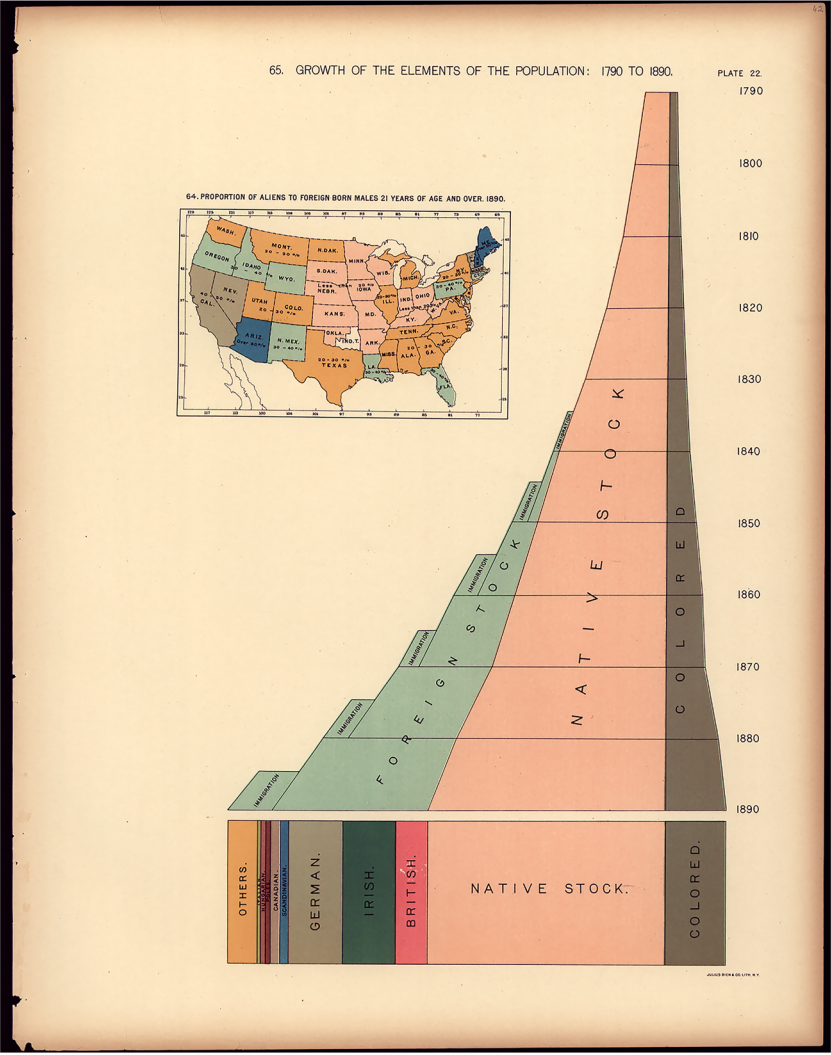
Inhabitants in continental United States [the “native stock”] in 1920’ does not include 1) immigrants from the [Western Hemisphere] or their descendants, 2) aliens ineligible to citizenship or their descendants 3) the descendants of slave immigrants, or 4) the descendants of the American aborigines. (as cited. in Ngai, 2006: 142).

This definition of American nationality excludes all Mexicans (“immigrants from the [Western Hemisphere] or their descendants,” all Chinese and Japanese (“aliens ineligible to citizenship or their descendants”), all Blacks (“the descendants of slave immigrants”), and Native Americans (“the descendants of the American aborigines”). With the non-native others identified, the “natives” of the United States were recognized as European Whites and their descendants.

The Immigration Act of 1924 was issued based on census data collected in the previous several decades. The *Statistical Atlases* visualized and perhaps shaped the exact definition of the native identity in the Immigration Act of 1924. As can be seen in Figure 1, “Growth of the Elements of the Population 1790–1890” in the 1898 *Atlas*, the population is divided into three major groups: “foreign stock,” “native stock” and “colored.” The “foreign stock” is further divided into major European nativities and the “others.”

The “native stock”—the one that represents American nativity—is not represented as all persons who were born in the United States, but as persons who were descended from Whites who came before 1790. The “native stock” excludes “colored,” and “foreign stock,” who were newer immigrants coming in after the 1830s and their descendants with a large portion being eastern and southern Europeans, and other non-White immigrants. The representation of the “native stock” constructs an American identity of being White and being northern and western European. Created in the late 19th century, this graph reflected and reinforced the concerns of the dominating “natives” group that finally led to the Immigration Act of 1924.

While the “native” population is often represented as a whole, creating a pan-ethnic native identity, the “foreign” population are subjected to further classification in the *Atlases*. In Figure 1, the nationalities of the “foreign stock” are specified for the audience for further inspection. Specifying the nationalities places a scrutinizing gaze on the nationalities or ethnicities of the new immigrants. The inspection on the foreign stock is in each volume of the *Atlases* from 1883 to 1924. Similar sets of pie charts were used to represent the constituent elements of the total population. As can be seen in Figure 2, the set of pie charts shows the total population from 1790 to 1890 in the 1898 *Atlas*.



(a)



(b)

Figure 1. “growth of the elements of the population 1790 to 1890” (Gannett, 1989 atlas, plate 22) (Note: The color version is available online).

The classification of the total population changes through the years. From 1790 to 1840, the population was classified into two mutually exclusive groups: “the White” and “the colored.” In the charts of 1850 and 1860, the “foreign White” is separated from the “White” category. In the charts after 1860, “native White with foreign parents” are further separated from the “native White” category. These added categories protect the integrity of the “native” Americans by identifying and setting the

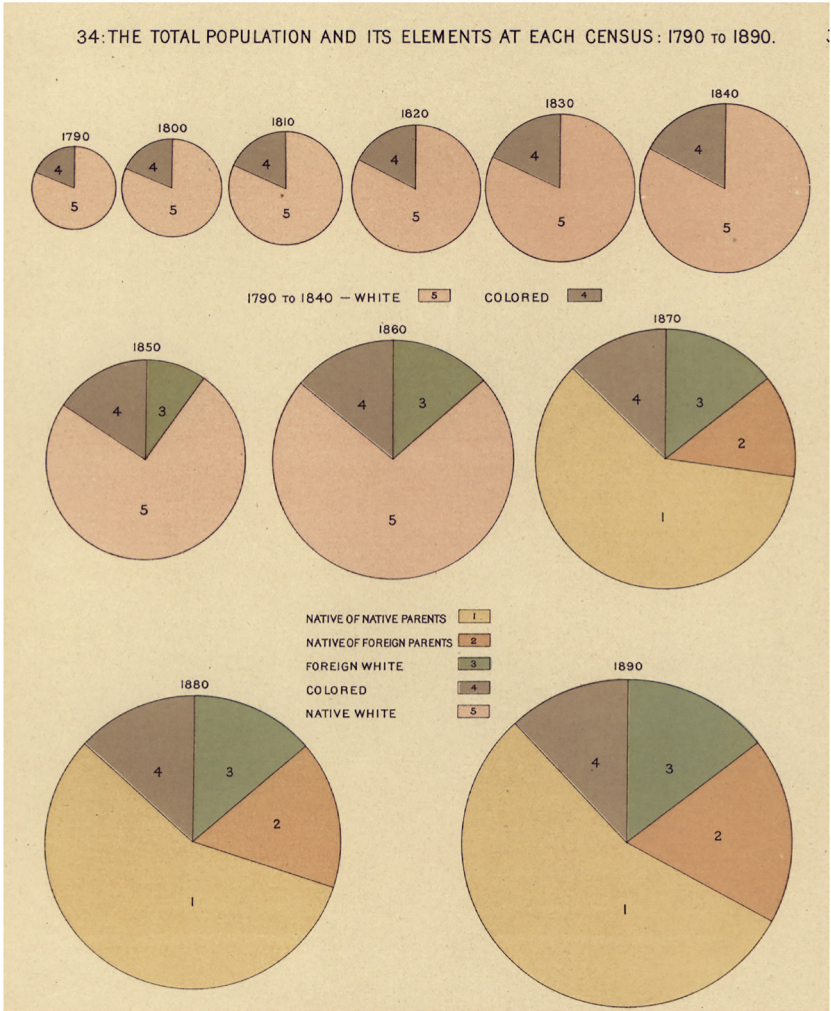


Figure 2. “The total population and elements from 1790 to 1890” (Gannett, 1898 atlas, plate 14) (Note: The color version is available online.).

foreign elements apart from the “native White” category. This classification visualizes the changing concerns of the nation from the Black population before the Civil War to foreign immigrants in the late 19th century. While maintaining the divide between White and Black, a hardened division is created between old immigrants and new immigrants by separating “foreign White” from “native White,” and “native White with foreign parents” from “native with native parents.” This visual division reflects a growing xenophobic sentiment towards immigrants in 19th century America.

The arrival of the new immigrants in the late 19th century brought a threat not only to the labor market but also to the cultural and ethnic identities of the Anglo-Saxon White immigrants. The contrastive classification of “native” versus “foreign” magnifies the ethnic differences between earlier Anglo-Saxon immigrants and new ones who were from southern and eastern Europe and from Asia. The ethnic differences between Anglo-Saxons and others were further magnified into biological differences in the *Atlases* by establishing a scientific association of specific diseases with the foreign population. In a line chart in the 1874 *Atlas*, Walker shows correlation between diseases and race and nationality (see Figure 3).

The “foreign” and “native” population were two major categories visualized in the chart. Zigzag lines go across the page connecting a variety of diseases to the “native” or “foreign” population. Through examining this correlation, Walker suggests that the foreign population has a “decided liability” to some diseases such as Bright’s disease, cancers, bronchitis, small-pox, diarrhea, and also is comparably immune from paralysis, rheumatism, and hydrothorax (Walker, 1874). The “very distinct predisposition” of the foreign population to certain forms of disease racializes national differences into inherited and immutable biological differences. The “foreign” and “colored” were therefore constructed as counterparts of each other.

The defined “others” (the “foreign” and the “colored”) are compared and contrasted with the “native Whites” creating a visible divide between “us” and “them” in the data

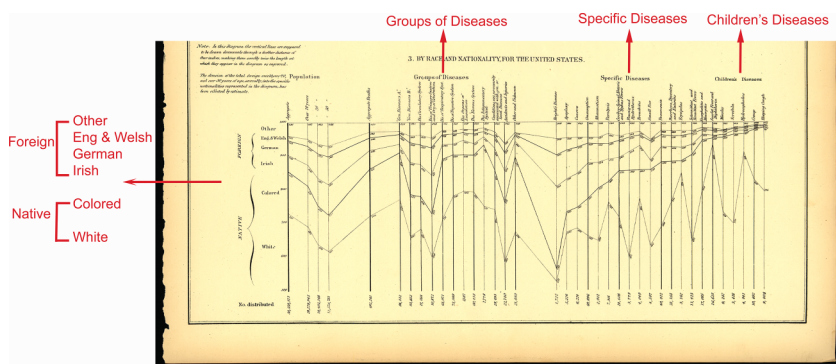


Figure 3. Disease chart “By race and nationality for the United States” (Walker, 1870 atlas, plate 44) (Red text is enlarged labels of the original chart added by the author) (Note: The color version is available online.).

displays of the *Statistical Atlases*. The divide is often seen in sets of opposites in the graphs of the national population such as the “White” versus “colored,” and the “native” versus “foreign.” The naming and classifying of the population provide a basis for graphical coding of these categories in the *Atlases*.

Coding the “Others” Graphically

After naming and classifying the “other,” the *Atlases* graphically coded them. Data displays rationalize population traits through their visual coding including coloring, shading, and spatial placements. Specific visual characteristics of a data display determine how readers interpret the meaning. Creators and readers with different cultural backgrounds may use and interpret visual codes (e.g. color, organization) differently. However, since the primary audiences of the *Atlases* were educated Americans in the 19th and 20th century, most of whom were middle-class European Whites, the discussion of graphic coding in this study is generally based on western cultural conventions. In the followings, I will discuss how coloring, shading, placement, and grouping can be used in racializing and marginalizing the other. Then I will illustrate the practice of these strategies in a population graph from the 1898 *Atlas*.

Racializing the “Others” by Coloring

Color is a key design element that enriches the meaning of a visual. The use of coloring can impose symbolic layers of meaning to a visual representation and influence the interpretation of the visual (Schenider and Nocke, 2018, p. 290). Color properties, such as hue and value, are often associated with certain meaning or emotions in a specific culture. A hue is a perceptible color on a color wheel, such as red, blue and green. The value of a color is the relative lightness and darkness of the color.

Hue. In the *Statistical Atlases*, a consistent use of hues is adopted to create visual references to racial and ethnic identities. The color black is used to represent the Black population; yellow is used for Chinese and Japanese population; and brown is used for Indian population. We can see from Figure 4 that the skin colors are assigned to particular minority groups in the graph.

The use of skin colors to reference races and ethnicities creates a visual synecdoche by using part of a thing (e.g. skin color) to represent the whole. Despite the many factors used to categorize race, skin colors have been a symbolic and racist method for identifying race.

Value. The value, darkness and lightness of the colors, is usually associated with certain emotions. As part of the 19th Victorian aesthetics and even that of today, bright colors often bring a sense of excitement, and darker colors can bring depressing emotions. The dark-light color association is examined by Kimball in his study on London poverty maps created in the same era as the *Atlases*. Kimball (2006)

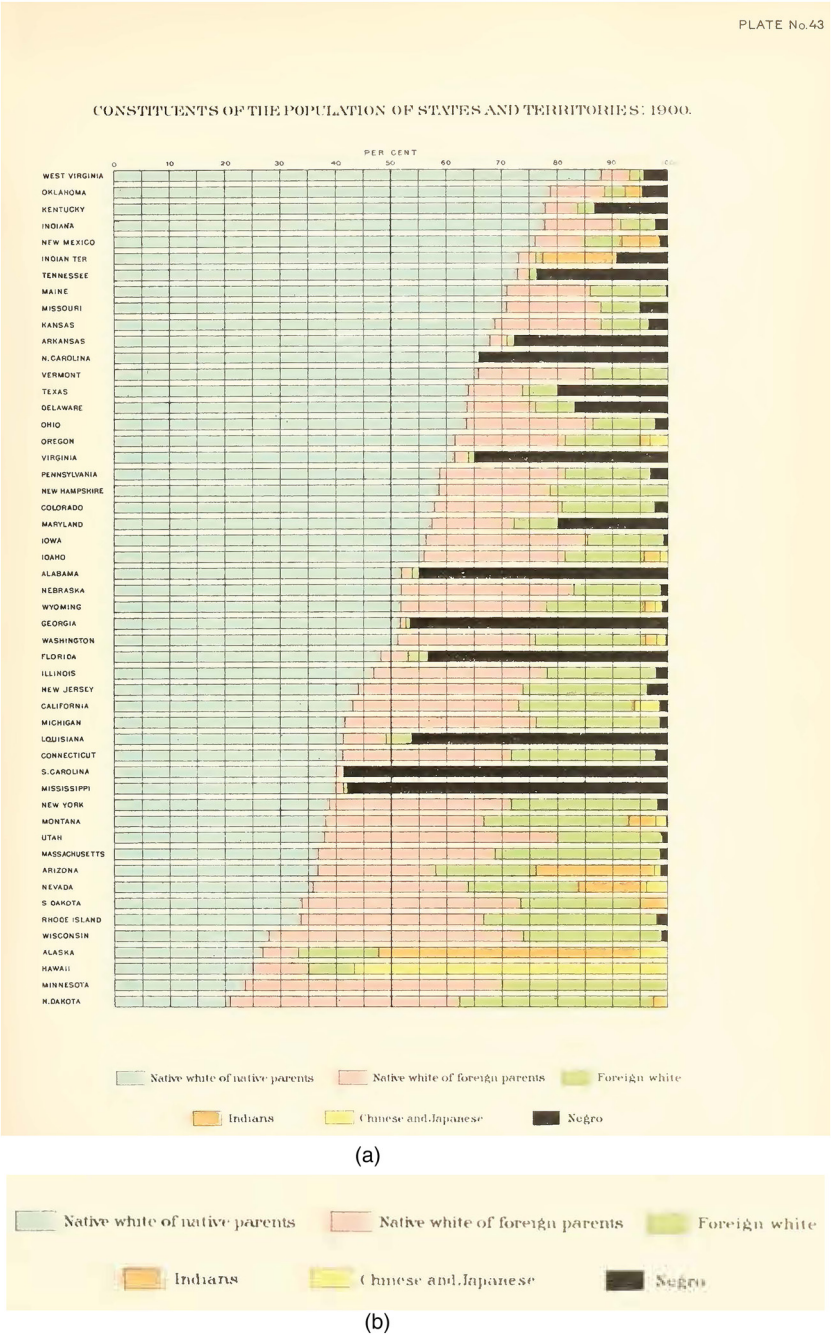


Figure 4. “constituents of the population of states and territories: 1900” (Gannett, 1903 atlas plate 43) (Note: The color version is available online.).

pointed out that “the color coding is the essential feature of the maps’ visual rhetoric” (p. 362). For example, in the London poverty maps, dark gradation coloring was used to represent poverty of London. He points out that “The color scale of the map represents poverty as a dark necrosis or cancer upon the body of London, with its tissues colored in healthy, flesh and blood-based tones” (Kimball, 2006, p. 360). The same kind of color rhetoric functions in the *Atlases*. Darker colors or shading (e.g. black, dark red) are generally used for the “others” (foreign population and non-Whites); while lighter colors (e.g. pink and White) are used for the “native” and “White” population, except for skin color references (e.g. yellow for Chinese).

Gradation coloring is used in the *Atlases* to mark the progressive growth of the foreign and Black population. As can be seen in Figure 5 in the 1874 *Atlas*, gradation coloring is used to represent the proportion of “foreign” and the “colored” population to the general population. The more concentrated areas of foreigners and the Blacks, the darker the shading is used. The gradation coloring creates a figure-ground contrast bringing attention to the “others” (foreigners and Blacks) as a concern of the nation.

Constructing Racial Hierarchy Through Spatial Placement

Layout is how visual elements are arranged in a design. Information designers seek to use layout to create visual hierarchy to guide the audience to the more important,

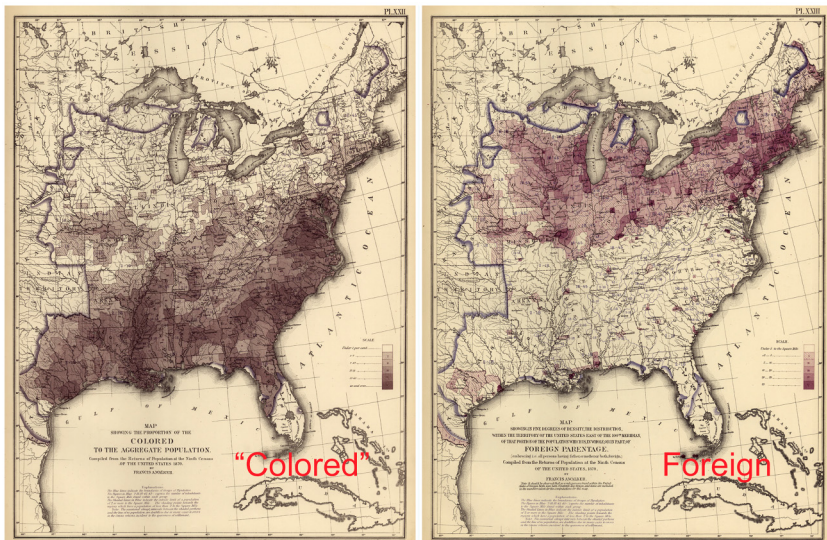


Figure 5. Left: “Map showing the proportion of the colored to the aggregate population” (Walker, 1874 atlas, plates 21). Right: “Map showing five degree of density, the distribution, within the territory of the United States east of the 100th meridian, of that proportion of the population which is, in whole, or in part, of foreign parentage” (Walker, 1874 Atlas, Plates 22) (Note: The color version is available online.).

urgent, and powerful information in the design. Spatial placement of a visual object can help to imply meaning and show relative status of a visual object among other objects. If something is particularly noticeable in the layout, we may assume that it is important because it is being emphasized.

Perceptually, the top and center of a layout are the most noticeable positions in a design. If an object is placed at the top, it may imply power and privilege over other objects. Similarly, if it is low on the page, it may be showing the object of an unfavorable social status. If an object is centered in the layout, this object is shown as important or advantaged in some way. If the object is on the fringe, its status can be weakened and shown as disadvantaged. Besides, since English readers tend to read from left to right, objects placed on the left are usually viewed as more important than the ones placed on the right.

The placement of visual objects in maps was critiqued by Barton and Barton. They point out that “the placement of visual elements becomes a way of imparting privilege” (Barton and Barton, 1993/2004, p. 236). Positioning to privilege includes centering, placing at the top, and ordering (pp. 236–237). In the *Atlases*, the hierarchical placement is apparent. The White and “native” population is mostly positioned in the center, top, and left, while the “others” are placed at the margin, bottom, and the right.

For example, in the two-page bar chart collection, “*Distribution of wage earners of the specified*”, the “native Whites” are given the most significant and powerful position of the top right on the left page, and the “colored” is found at the least important position of the bottom left on the right page (see Figure 6).

Another example can be seen in Figure 7, *Distribution of those engaged in certain selected occupations by color and nationality* in the 1898 *Atlas*, the populations are ordered from left to right as 1. “native White, native parents,” 2. “native White,” “foreign parents;” 3. “foreign White;” 4. “colored.” The hierarchy is constructed in an ordered list from those with the highest status (native White, native parents) to lower status (“foreign White”) and to the lowest (African Americans).

Within the “foreign White” category, an ethnical hierarchy is also created in a numbered order from left to right: 1. Irish, 2. English, Welsh and Scotch, 3. Germans, 4. Scandinavians 5. Canadians, 6. French, 7. Italians, 8. Slavs, 9. Other foreign. Immigrants from western Europe such as “Irish” “English, Welsh and Scotch” and “Germans” are placed at more favorable positions in the list than the southern Europeans of Italians and Slavs. Other non-White foreigners are placed at the bottom of the ethnic hierarchy. However, one exception of color coding is used here: warmer colors (pink, and light pink) are used for the foreign and “colored” while colder colors of green and blue are used for the “native White.”

The hierarchical ordering is not only racist but also gendered. In the *Illiteracy map of selected classes* the graph starts at the top right with “native White man” and ends at the bottom left with “colored females.” (see Figure 8), The dark red shading of the illiterate mapping shows “colored women” as the least literate population of the nation and hence constructs them as being the lowest in social status.

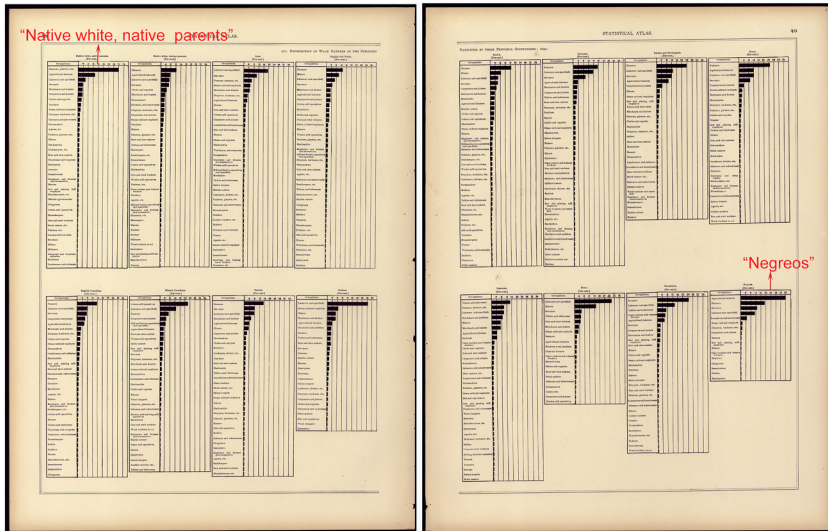


Figure 6. “distribution of wage earners of the specified nativities by their principal occupations: 1890” (Gannett, 1898 atlas, p. 48) (Red text is enlarged labels of the original chart added by the author). (Note: The color version is available online.).

Inspecting the “Others” by Rhetorical Grouping and Scattering

Visual grouping is another layout strategy that shows the overall structure of a design. The design principle of “proximity” is a powerful grouping tool. Proximity suggests that when items are positioned in close proximity, they appear to have a relationship or belong together (Williams, 2015). When visual objects are purposefully organized into groups, readers can make meaningful connections across the objects in a design. Similarly, if items were separated and placed far from each other, the relationship between the items diminishes. Shriver (2012) pointed out that grouping content spatially can help “make what otherwise might be invisible structures apparent to the readers” (p. 391).

Grouping. As discussed previously, population categories are rhetorical constructions. Grouping these categories in a design can construct a pan-ethnic identity for the groups or create visual divide between “us” and “them.” The “native” population and especially the “native White” is presented as one entity, while Blacks and the foreign population are grouped as counterparts of each other.

For example, in 1874 Atlas, the population density of the “foreign” and the “colored” were visualized on pages next to each other as a group complementing each other (see Figure 9). The gradation shading of the two maps emphasizes the counterpointed nature of the two: the “colored” concentrating in the south while foreigners

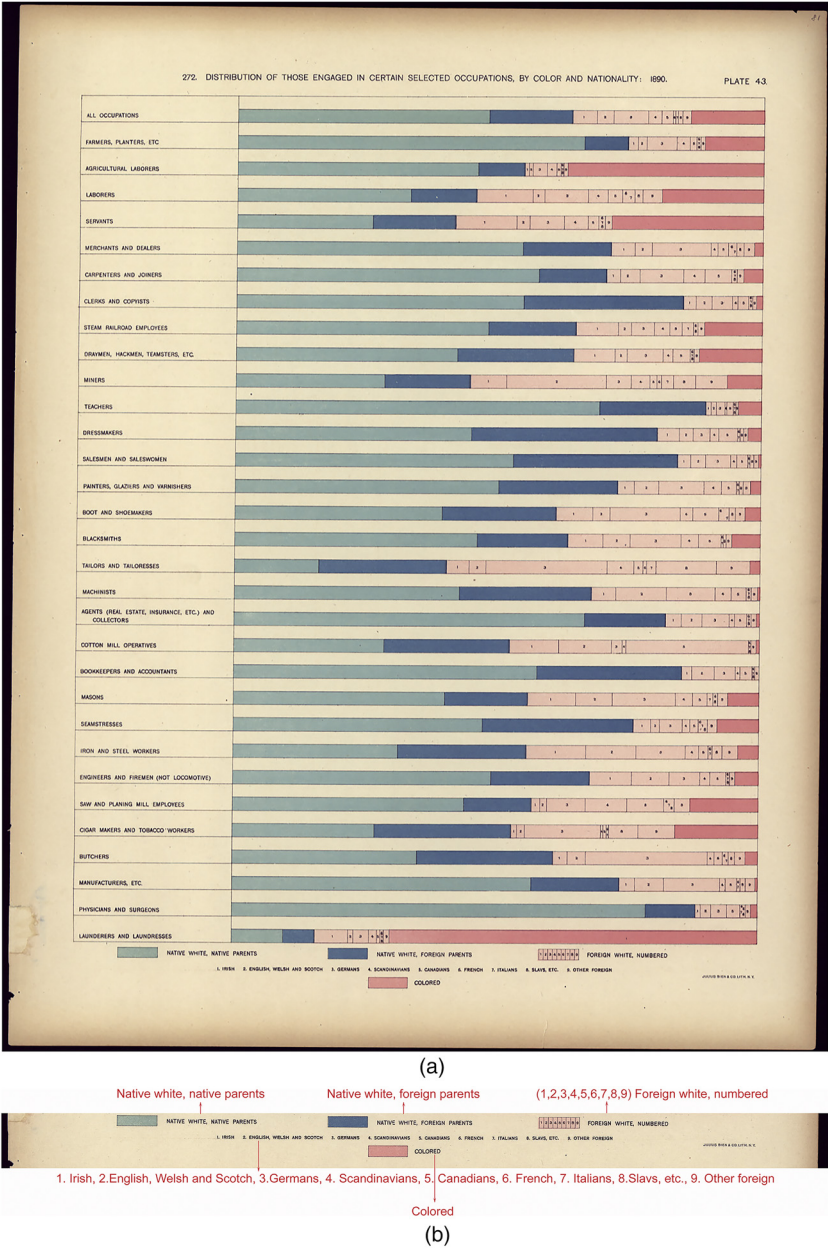


Figure 7. “distribution of those engaged in certain selected occupations, by color and nationality: 1890” (Gannett, 1898 atlas, plate 43) (Note: The color version is available online.).

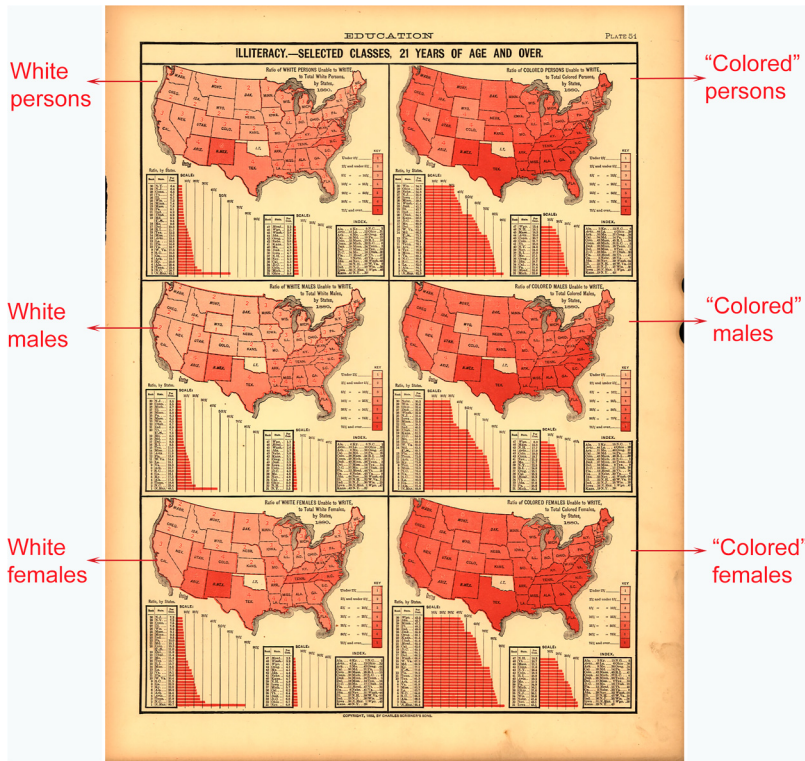


Figure 8. "illiteracy. Selected Classes, 21 years of age and over" (Hewes and Gannett, 1883 Atlas, Plate 51) (Note: The color version is available online).

clustering in the north. As Walker (1874) explained in the Preface of the *Atlas*, "speaking broadly, where the Blacks are found in the United States, the foreigners are not," "in a high degree complementary in their location" (p. 5).

Blacks' attachment to hot weather and certain latitude of location had been a popular racist assumption about Blacks in the 19th century. Walker states in his article, "The Colored Race in the United States" in 1891 that Blacks "represented a race bred under tropical conditions, and could move up the mountainside or go northward only at a large sacrifice of vitality and force" (p. 501). The immobility of a race is also a symbol of less productivity. Bonded to specific latitudes, foreigners and Blacks are constructed as being inferior races. In contrast with the two immobile races, native Whites are represented as being perfectly mobile and productive. Walker states that native Whites "represented a race bred in the northern latitudes, and was hence thoroughly at home on the mountain side or table-land; while yet, by the privilege of his strain he could, without danger or great inconvenience, move southward if his interests required." (as cited in Hannah, 2000, p. 179). Here, Walker not only emphasizes the

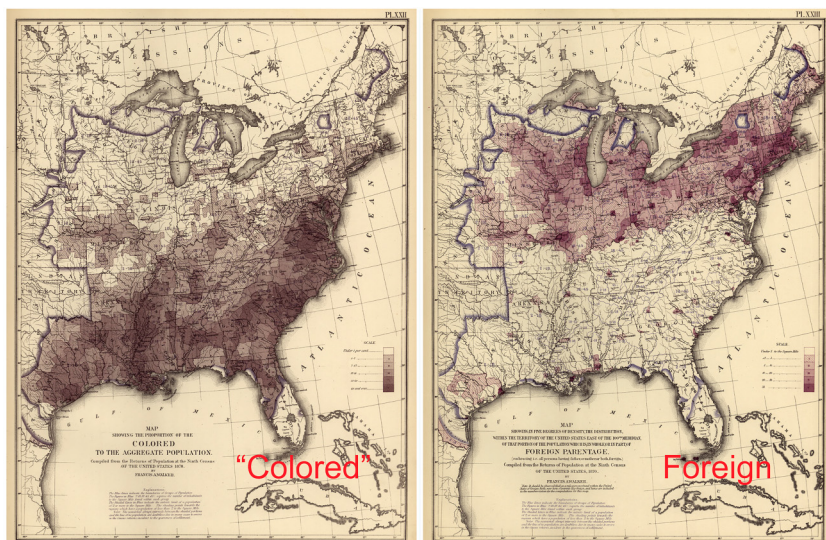


Figure 9. Left: “Map showing the proportion of the colored to the aggregate population” (Walker, 1874 atlas, plates 21). Right: “Map showing five degree of density, the distribution, within the territory of the United States east of the 100th meridian, of that proportion of the population which is, in whole, or in part, of foreign parentage” (Walker, 1874 Atlas, Plates 22) (Note: The color version is available online.).

superior nature of the native Whites but also “naturalized the spread of ‘native Americans’ over the whole continent.” (Hannah, 2000, p. 179).

As a complementary element of Blacks, foreigners are bonded to the northern states that the Blacks are not. Parallely grouping foreign immigrants with Blacks fosters the view that “foreigners” constitute a race. Takaki (2008) explains the racism behind this kind of parallelism: “What Whites did to one racial group had direct consequences for others, and Whites did not artificially view each group in a vacuum: rather in their minds, they lumped the different groups together or counterpointed them against each other” (p. xiv).

Another example can be seen in Figure 4, the chart label places all Whites on the first line while “Indians,” “Chinese,” and “Negros” were grouped together on the second line. Contrasted with the Whites, these minority groups were grouped and equated with each other as a less favorable group of three.

Separating, scattering, and normalizing. Opposite to grouping, the “foreign” population or newer immigrants—especially non-Whites and non-Europeans—are often represented as separated and sub-divided ethnic groups. Separation and subdivision of the ethnicities of foreign population made their ethnicities visible for the purpose of better surveillance and control. Graphs examining the conditions of the “foreign”

population make up the majority of population graphs in the *Atlases*. The growth, age, sex, nationality, illiteracy, linguistic capabilities, geographic distribution, and occupations of the foreign population are examined in various forms of charts. Through these graphs, the newcomers are rendered visible for public inspection and regulation.

Graphs such as Figure 10 on the composition of the foreign-born population are common in the *Atlases*. The foreign population is subdivided into 13 ethnic groups, and each ethnic group is scattered in the pie charts of the states. Scattering the population diminishes the presence of each ethnic group as a whole in the nation and thus diffuses the power of an individual ethnicity. The dispersed ethnical groups have fewer chances to form a stronger group identity and gain power. In addition, “Chinese” is singled out in the chart as the only non-White group. The identification of the Chinese population guides attention to the group and calls for examination of the group after the 1882 Chinese Exclusion act (see Li, 2020).

Diffusing and weakening the power of individual ethnicities provides a way to normalize the “other” population. In Figure 11, the age and gender composition of both “native population” and “foreign population” are represented in pyramid-shaped population distribution charts. The “foreign population” were scattered into the charts of each state. Visually, the graphs of the native population were mostly shaped in steady pyramids with the youngest age group comprised the largest percentage of the population. The age structure is regarded as projecting the healthy growth of the native population.

In contrast, the foreign population pyramids are non-standard pyramids with a bulge around the middle-age groups. Some of the graphs also unusually extend sideways to show unbalanced distribution of gender. The shape of the pyramids indicates the abnormality of the age structure. As human eyes tend to look for abnormality and outliers, these graphs not only allow for public examination of the foreign population but also highlight specific problems of the group as a result of the examination. The abnormality and peculiarity of the foreign population calls for management and normalization by the States.

A Prominent Example of Graphical Coding of the “Others”

The graphic coding of coloring, placement, and grouping can be collectively illustrated in the following graph (also discussed previously). *The Growth of the Elements of the Population (1790 to 1880)* chart in the 1898 *Atlas* illustrates the racialization and marginalization of minority groups and constructs an American national identity perceived in the late 19th century through coloring, placement, and grouping (Figure 12).

Coloring. The chart divides the population into three major groups: “foreign stock,” “native stock” and “colored.” A warmer and lighter color (pink) was assigned to the “native stock” while colder and darker colors (black and blue) are used for Black and foreign population respectively (color online). Black, the color of the black skin, labels the racial stigma. In the “foreign stock,” British immigrants are painted with a darker pink, a color similar to the color pink of the “native stock,” showing a close

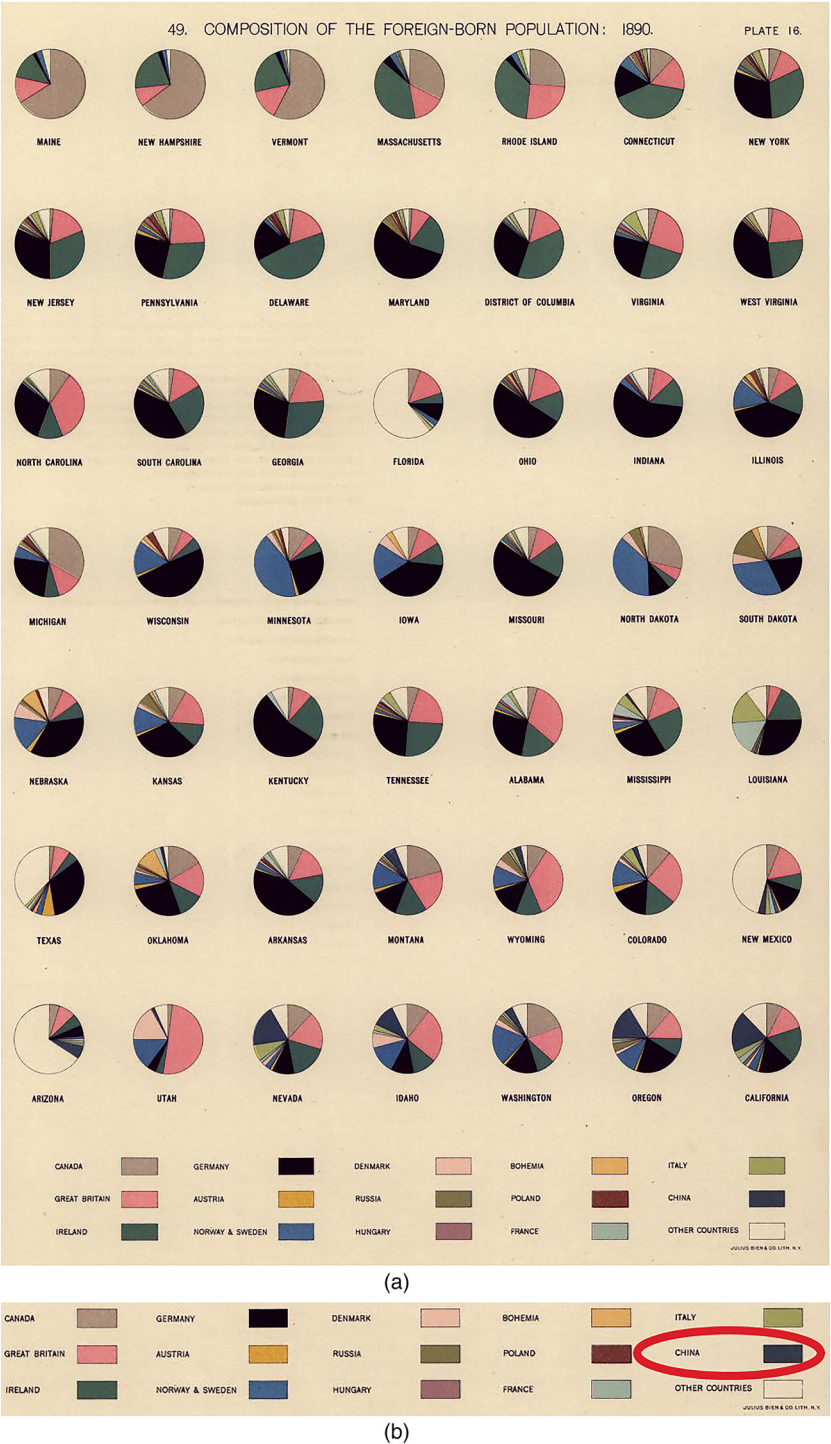


Figure 10. “The composition of the foreign born population: 1890” (Gannett, 1898 atlas plate 16)
(Note: The color version is available online.).

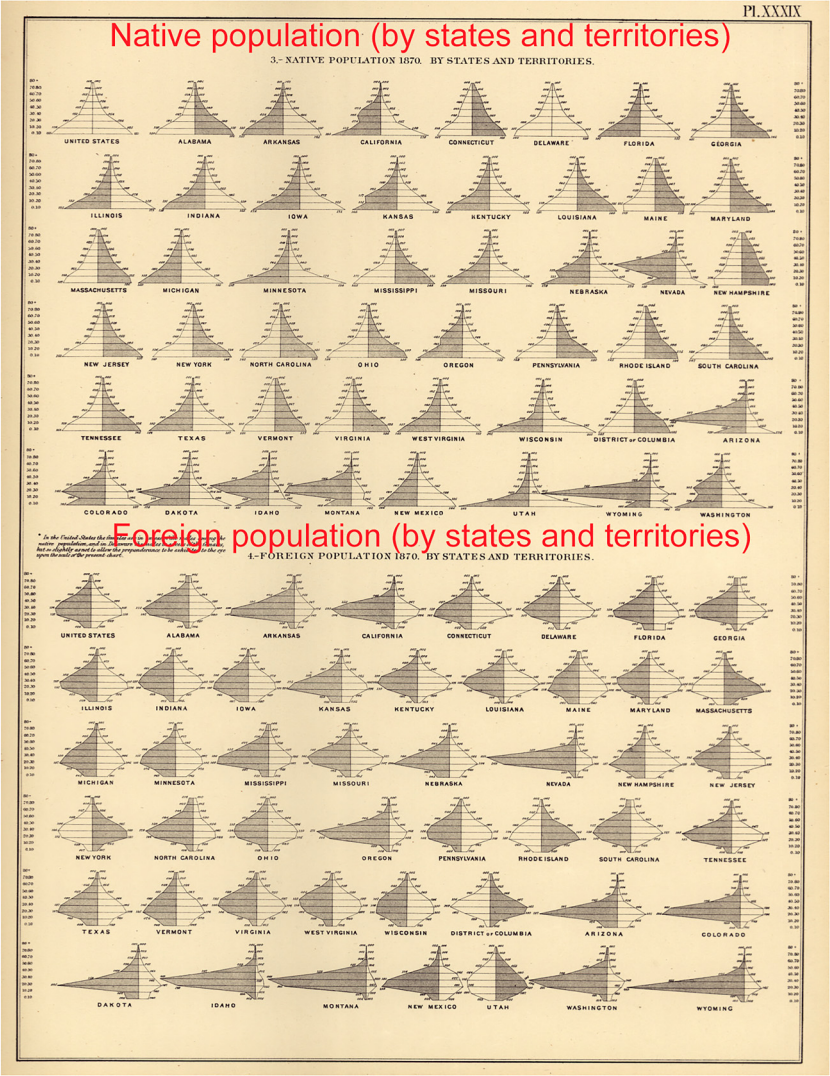


Figure 11. Age distribution: native population and foreign population. (Walker, 1874 Atlas, Plate XXXIX) (Note: The color version is available online).

relationship to the American “natives” (color online). Yellow is used for the unidentified “other” of the foreign population. Because of the 1882 Chinese Exclusion Act, all immigrants from Asia are invisible in the chart except for a very small number of Japanese and Filipinos included in the “other” category. The color yellow, the assumed skin color of Asians, highlights the biological traits of the Asian identity (color online).

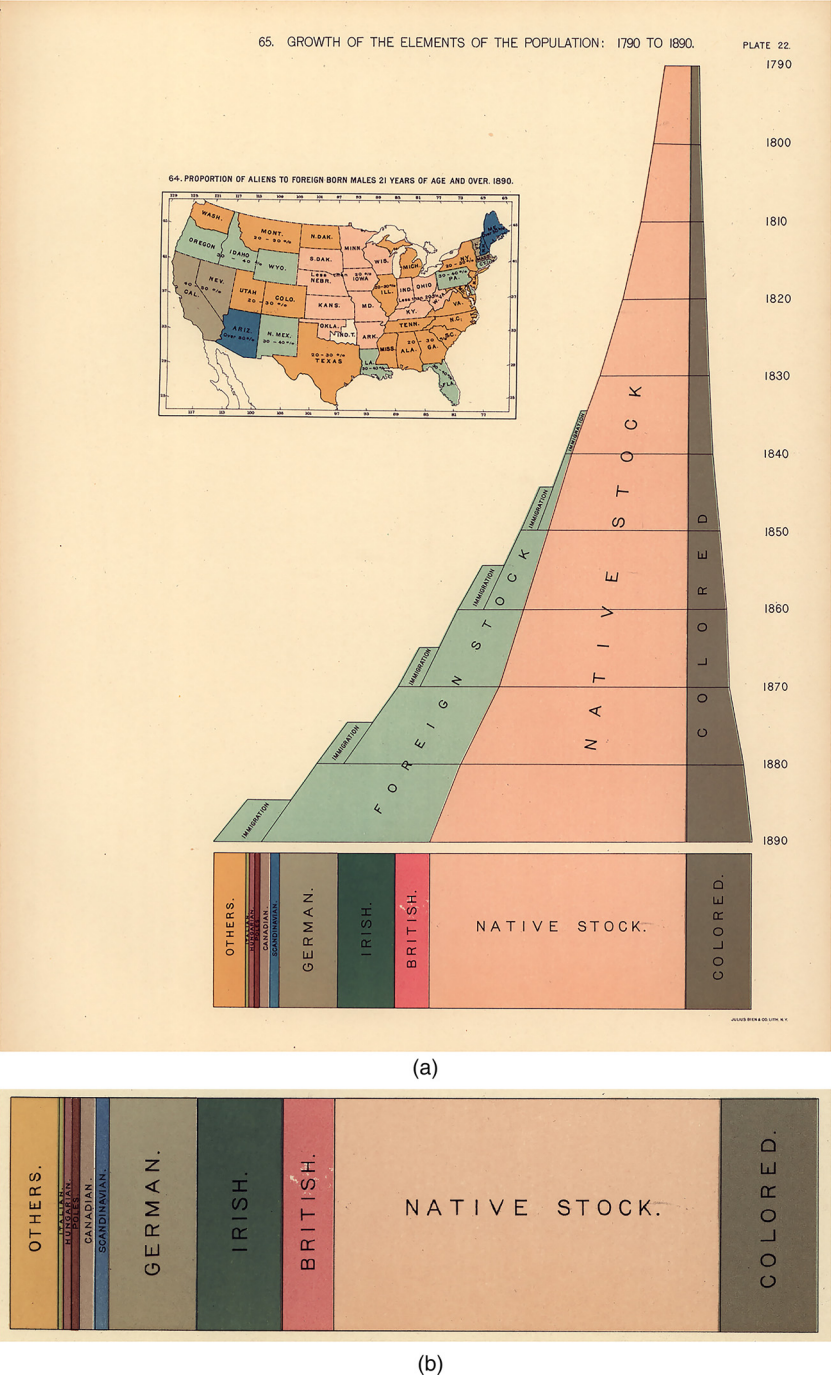


Figure 12. “growth of the elements of the population 1790 to 1890” (Gannett, 1989 atlas, plate 22) (Note: The color version is available online.).

Placement. The “native stock” is placed at the center of the population groups forming the focus of the view. The foreign population and African Americans are pushed to the left and right margins, respectively. Placed at the two respective sides of the “native stock,” foreign and Black population are constructed as the counterpart of each other, both of which are undesirable “others,” threatening the existence of the “native stock.”

With a striking visual effect, the graph shows the rapid growth of the population of the “foreign stock” from zero to a size comparable to the “native stock” since the 1840s. The dramatic size change of the “foreign stock” stream creates a visual shock for the audience and calls for attention to examine and manage the “foreign” population.

Within the “foreign stock” section, the seats for foreign immigrants are arranged hierarchically according to their ethnicities. From the center to the margin, we have British, Irish, German, Scandinavian, Canadian, Polish, Hungarian, Italian, and “others,” a ranking order based on the supposedly valued characteristics of the ethnicities. Western Europeans (British, Irish, German) are offered privileged seats close to the center and next to the “native stock.” At the far margins of the chart is the “other” category, most of which are non-Europeans and non-Whites. This visual structure accurately predicted the hierarchical assignment of annual immigration quotas in the Immigration Act of 1924. As discussed previously, this Act assigned the largest quotas to British, Germans, and Scandinavians, while Italians, Greeks, Slavs, and Jews got only a very small share (Purcell, 1995). The small number of non-European immigrants are represented as unnamed and faceless “others” in the chart.

Grouping. The three major population categories in the chart show a divide between “us” the “native stock,” and “them,” the “foreign” and the “colored.” While older Western European immigrants are represented as one entity of the “native stock,” the foreign population is subdivided into major European nativities and “others” for further examination.

In the foreign group, western European ethnicities (British, Irish, German) are closely grouped together. The eastern and southern (Italians, Hungarians, Poles, Scandinavians) Europeans are placed next to each other. The proximity grouping emphasizes the relationship of the two subgroups of foreigners and their respective relationship to the “native stock.”

The “other” category in the foreign stock groups the least powerful immigrants together including immigrants from Asia and other non-Whites. These minority groups aren’t even identified but are collectively put in an unnamed category. Blending the powerless groups together erases the individual identity of an ethnic group and diminishes its power.

In sum, through naming and categorizing the population groups, and through the use of graphic coding of the visual elements, the chart privileges and empowers the

“native White” population while marginalizes and racializes the foreign and Black population in the nation. The chart imagines a national identity of being White and having European origins. Designed in the late 19th century, the chart reflects and shapes the political and racial discourse in the era.

Conclusion

This study answers the call for the redress of racism and xenophobia in the technical communication field through the analysis of the visual strategies in the *Statistical Atlases* in the late 19th and early 20th century. By looking at the historical data visualization of race and ethnicities in the *Atlases*, I provide “insights about how, why, and in what way certain populations have been oppressed and map historical contexts to contemporary situations” (Jones, 2016, p. 351). Along with Scott et al.’s (2006) call, the study also intended to “recognize[s] technical texts as connected to broader cultural practices, as always-already ideological, and enmeshed in forms of power” (p. 5).

I argue that the process of taking census and visualizing census data for the public is fraught with political ideology of racism and xenophobia of the historical era. The initiator of the *Statistical Atlases*, Francis Walker, and his followers brought the prevailing ideology of xenophobia and racism of the 19th century into the design of the data graphs. White supremacy ideology was sustained and promoted through the use of particular design strategies in the *Atlases*. The *Atlases* constructed a demarcation between “us”—the Western European Whites and “them”—the Black and the new immigrants. The population categories are ranked in a hierarchy of desirability with the White Anglo-Saxons at the top and Black and other foreign population at the bottom.

This hierarchical representation is constructed through the rhetorical strategies of naming, categorization, and graphical coding strategies, including the use of coloring, spatial placement, and strategic grouping and scattering.

Naming and Categorization

Through inclusive, exclusive, and contrastive grouping, the naming and categorization of population groups identify the “others,” and create a hardened division between “us” and “them.” Presented as “exact knowledge,” the *Atlases* construct the classification as objective and naturalized categories of population upon which readers may not doubt or reflect.

Coloring

The use of perceived skin colors to represent races in the data graphics legitimizes social constructs of racial differences and hierarchies. The consistent use of darker colors or shading for African Americans and new immigrants creates negative associations with minority groups and problems. These color strategies demonstrate racism, colorism, and develop implicit bias.

Spatial placement

Spatial placement of visual objects emphasizes the status of the object in relation to other objects in a design. In the *Atlases*, the dominate group (White and “native” population) is placed in powerful positions: center, top, and left. The “others” (African Americans and new immigrants) are placed at weakened positions: the margin, bottom, and the right. An ethnical hierarchy is also constructed through ordered lists from left to right.

Grouping and scattering

Rhetorical grouping of visual objects creates relationships between the objects. African Americans and the foreign population are grouped and contrasted as counterparts of each other. While western European Whites are represented as one entity, non-Whites and non-Europeans are often represented as separated and sub-divided ethnic groups. Separation and subdivision of the ethnicities of foreign population made their ethnicities visible for the purpose of surveillance and control. Scattering the groups in the graphs weakens the power of individual ethnicities and provides a way to normalize the “other” population.

In general, while the “native White” is empowered through bright coloring, centering, and topping, the foreigners and Blacks are disadvantaged through dark shading, marginalization, purposeful grouping and scattering.

Through the study, I call for a critical awareness on how big data and data visualization could be used to reflect and reshape the history and can be used to influence political sentiment of immigration and racial disputes today. The visual strategy framework synthesized in the paper can be used as an analysis method on the discriminative language of data visualization in general. The study also informs technical communicators of their own rhetorical strategies in today’s era of big data.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

ORCID iD

Li Li  <https://orcid.org/0000-0002-2522-6941>

Note

1. “Native white” is a term used in the *Statistical Atlases*. “Native” here does not refer to native Americans/indigenous people in today’s sense. The indigenous population had not officially been covered in the census until 1890. The representation of indigenous population is not in the scope of this study. I am aware of the problematic and racist connotation of the term. Other similar racist terms are also used in the *Atlases* such as “colored,” and “negro.” I keep these terms in the article so that readers can see the original versions used in the *Atlases*. I use them in quotation marks to show that those are not my language and I disapprove using these terms.

References

- Agboka, G. Y. (2013). Participatory localization: A social justice approach to navigating unenfranchised/disenfranchised cultural sites. *Technical Communication Quarterly*, 22(1), 28–49. <https://doi.org/10.1080.10572252.2013.730966>.
- Agboka, G. Y. (2014). Decolonial methodologies: social justice perspectives in intercultural technical communication research. *Journal of Technical Writing and Communication*, 44(3), 297–327. <https://doi.org/10.2190/TW.44.3.e>
- Anderson, B. (2016). *Imagined communities : Reflections on the origin and spread of nationalism*. (Rev. and extended). Verso.
- Anderson, M. J., & Fienberg, S. E. (1999). *Who counts? The politics of census-taking in contemporary America*. Russell Sage Foundation.
- Balzhiser, D., Pimentel, C., & Scott, A. (2019). Matters of form: Questions of race, identity, design, and the u.s. Census. *Technical Communication Quarterly*, 28(1), 3–20. <https://doi.org/10.1080/10572252.2018.1539192>.
- Banks, A. (2006). *Race, rhetoric, and technology: Searching for higher ground*. Lawrence Erlbaum.
- Barton, B., & Barton, M. (2004). Ideology and the map: Toward a postmodern visual design practice. In J. Johnson-Eilola, & S. A. Selber (Eds.), *Central works in technical communication* (pp. 232–252). Oxford University Press. (Original work published in 1993).
- Bertin, J. (2001). Matrix theory of graphics. *Information Design Journal*, 10, 5–19. <https://doi.org/10.1075/idj.10.1.04ber>.
- Blyler, N. (1998). Taking a political turn: The critical perspective and research in professional communication. *Technical Communication Quarterly*, 7(1), 33–33. <https://doi.org/10.1080/10572259809364616>
- Brasseur, L. E. (2005). Florence nightingale’s visual rhetoric in the rose diagrams. *Technical Communication Quarterly*, 14(2), 161–182. https://doi.org/10.1207/s15427625tcq1402_3.
- Cleveland, W. S., & McGill, R. (1987). Graphical perception: The visual decoding of quantitative information on graphical displays of data. *Journal of the Royal Statistical Society Series A (General)*, 150(3), 192–229. <https://doi.org/10.2307/2981473>.
- Colton, J. S., & Holmes, S. (2018). A social justice theory of active equality for technical communication. *Journal of Technical Writing and Communication*, 48(1), 4–30. <https://doi.org/10.1177/0047281616647803>.
- Daniels, R. (2002). *Coming to America : A history of immigration and ethnicity in American life* (2nd ed). Perennial.

- Friendly, M. (2008). A brief history of data visualization. In C. Chen, W. K. Härdle, & A. Unwin (Eds.), *Handbook of data visualization* (pp. 15–57). Springer.
- Frith, Jordan. (2017). Big data, technical communication, and the smart city. *Journal of Business and Technical Communication*, 31(2), 168–187. <https://doi.org/10.1214/08-ST5268>.
- Funkhouser, H. G. (1937). Historical development of the graphical representation of statistical data. *Osiris*, 3, 269–404. <https://doi.org/10.1086/368480>
- Gannett, H. (Comp.). (1898). *Statistical atlas of the United States, based upon results of the eleventh census*. U.S. Census Office. GPO.
- Gannett, H. (Comp.). (1903). *Statistical atlas: twelfth census of the United States, taken in the year 1900*. U.S. Census Office. U.S. Census Office.
- Gannett, H., & Hewes, F. W. (1883). *Scribner's statistical atlas of the United States showing by graphic methods their present condition and their political, social and industrial development*. Charles Scribner's Sons.
- Gossett, T. F. (1997). *Race : The history of an idea in America*. Oxford University Press. <https://public.ebookcentral.proquest.com/choice/publicfullrecord.aspx?p=241567>.
- Graham, S., Kim, S., Devasto, D. M., & Keith, W. (2015). Statistical genre analysis: Toward big data methodologies in technical communication. *Technical Communication Quarterly*, 24, 70–104. <https://doi.org/10.1080/10572252.2015.975955>.
- Haas, A. M. (2012). Race, rhetoric, and technology: A case study of decolonial technical communication theory, methodology, and pedagogy. *Journal of Business and Technical Communication*, 26(3), 277–277. <https://doi.org/10.1177/1050651912439539>
- Haas, A. M. (2020, June 26). *ATTW President's call to action*. Association of Teachers of Technical Writing. <https://attw.org/blog/attw-presidents-call-to-action/>.
- Haas, A. M., ... (2021, March 26). *ATTW Statement against anti-asian racism and call to action*. Association of Teachers of Technical Writing. <https://attw.org/blog/attw-statement-against-anti-asian-racism-and-call-to-action/>.
- Hannah, M. G. (2000). *Governmentality and the mastery of territory*. The Cambridge University Press.
- Hannah, M. G. (2001). Sampling and the politics of representation in US census 2000. *Environment and Planning D: Society and Space*, 19, 515–534. <https://doi.org/10.1068/d289>
- Hepworth, K., & Church, C. (2018). Racism in the machine: Visualization ethics in digital humanities projects. *Digital Humanities Quarterly*, 12(4), n.p. <http://www.digitalhumanities.org/dhq/vol/12/4/000408/000408.html>.
- Herndl, C. G. (1993). Teaching discourse and reproducing culture: A critique of research and pedagogy in professional and non-academic writing. *College Composition and Communication*, 44, 349–363. <https://doi.org/10.2307/358988>
- Johnson, J. R., Pimentel, O., & Pimentel, C. (2008). Writing New Mexico white: A critical analysis of early representations of New Mexico in technical writing. *Journal of Business and Technical Communication*, 22, 211–236. <https://doi.org/10.1177/1050651907311928>
- Jones, J. (2015). Information graphics and intuition: heuristics as a techne for visualization. *Journal of Business and Technical Communication*, 29(3), 284–313. <https://doi.org/10.1177/1050651915573943>.
- Jones, N. N. (2016). The technical communicator as advocate: Integrating a social justice approach in technical communication. *Journal of Technical Writing and Communication*, 46(3), 342–361. <https://doi.org/10.1177/0047281616639472>

- Jones, N. N., & Walton, R. (2016). Using narratives to foster critical thinking about diversity and social justice. In M. Eble, & A. Haas (Eds.), *Key theoretical frameworks: teaching technical communication in the Twenty-First Century*. (pp. 241–267). Utah State University Press.
- Katz, S. B. (1992). The ethic of expediency: Classical rhetoric, technology, and the holocaust. *College English*, 54(3), 255–255. <https://doi.org/10.2307/378062>
- Katz, S. B. (1993). Aristotle's rhetoric, hitler's program, and the ideological problem of praxis, power, and professional discourse. *Journal of Business and Technical Communication*, 7(1), 37–62. <https://doi.org/10.1177/1050651993007001003>
- Kertzer, D. I., & Arel, D. (2002). *Census and identity : The politics of race, ethnicity, and language in national census* (Ser. New perspectives on anthropological and social demography). Cambridge University Press.
- Kimball, M. A. (2006). London Through rose-colored graphics: visual rhetoric and information graphic design in Charles booth's maps of London poverty. *Journal of Technical Writing and Communication*, 36(4), 351–381. <https://doi.org/10.2190/K561-40P2-5422-PTG2>.
- Kinross, R. (1989). The rhetoric of neutrality. In V. Margolin (Ed.), *Design discourse: history/theory/criticism* (pp. 131–143). U of Chicago P.
- Kostelnick, C. (2004). Melting-pot ideology, modernist aesthetics, and the emergence of graphical conventions: The statistical atlases of the United States, 1874–1925. In C. A. Hill, & M. Helmers (Eds.), *Defining visual rhetorics* (pp. 252–242). Lawrence Erlbaum.
- Kostelnick, C. (2016). Mosaics, culture, and rhetorical resiliency: The convoluted genealogy of a data display genre. In M. A. Kimball, & C. Kostelnick (Eds.), *Visible numbers: essays on the history of statistical graphics* (pp. 177–206). Ashgate Publishing Company.
- Kostelnick, C. (2017). Social and cultural aspects of visual conventions in information design. *Information Design Research and Practice*, 257–273.
- Kostelnick, C., & Hassett, M. (2003). *Shaping information: The rhetoric of visual conventions*. Southern Illinois University Press.
- Li, L. (2020). Visualizing Chinese immigrants in the U.S. Statistical atlases: A case study in charting and mapping the other(s). *Technical Communication Quarterly*, 29(1), 1–17. <https://doi.org/10.1080/10572252.2019.1690695>.
- Longo, B. (1998). An approach for applying cultural study theory to technical writing research. *Technical Communication Quarterly*, 7, 53–73. <https://doi.org/10.1080/10572259809364617>
- Mezey, N. (2002). Erasure and recognition: The census race and the national imagination. *Northwestern University Law Review*, 97(4), 1701–1768.
- Mirel, B. (2004). Introduction to “ideology and the map” by Barton and Barton. In J. Johnson-Eilola, & S. A. Selber (Eds.), *Central works in technical communication* (pp. 232–252). Oxford University Press.
- Monmonier, M. (1994). The rise of the national atlas. *Catographica*, 31(1), 1–15. <https://doi.org/10.3138/T3NN-QL75-753L-25G7>.
- Ngai, M. M. (2006). The immigration Act of 1924: inventing national origins. In K. E. Laura (Ed.), *Immigration* (pp. 139–151). T. Gale. (Reprinted from The Architecture of Race in American Immigration Law: A Reexamination of the Immigration Act of 1924. *Journal of American History*, 86, 1999).
- Nobles, M. (2000). *Shades of citizenship : race and the census in modern politics*. Stanford University Press.
- Pimentel, C., & Balzhiser, D. (2012). The double occupancy of hispanics: Counting race and ethnicity in the U.S. Census. *Journal of Business and Technical Communication*, 26(3), 311–311. <https://doi.org/10.1177/1050651912439696>

- Purcell, L. E. (1995). *Immigration* (Ser. Social issues in American history series). Oryx Press.
- Rawlins, J. D., & Wilson, G. D. (2014). Agency and interactive data displays: Internet graphics as co-created rhetorical spaces. *Technical Communication Quarterly*, 23, 303–322. <https://doi.org/10.1080/10572252.2014.942468>.
- Savage, G., & Mattson, K. (2011). Perceptions of racial and ethnic diversity in technical communication programs. *Programmatic Perspectives*, 3(1), 5–57.
- Schneider, B., & Nocke, T. (2018). The Feeling of Red and Blue—A Constructive Critique of Color Mapping in Visual Climate Change Communication. In: Leal Filho W., Manolas E., Azul A.
- Schor, P., & Schor, P. (2017). *Counting Americans : How the us census classified the nation* (L. A. Weiss, Trans.). Oxford University Press.
- Scott, J. B., & Longo, B. (2006). Guest editors' introduction: Making the cultural turn. *Technical Communication Quarterly*, 15(1), 3–7. https://doi.org/10.1207/s15427625tcq1501_2
- Scott, J. B., Longo, B., & Wills, K. V. (2006). *Critical power tools: technical communication and cultural studies*. SUNY Press.
- Shriver, K. (2012). What do technical communicators need to know about information design. In J. Johnson-Eilola, & S. A. Selber (Eds.), *Solving problems in technical communication* (pp. 386–427). University of Chicago Press.
- Slack, J. D., Miller, D. J., & Doak, J. (1993). The technical communicator as author: Meaning, power, authority. *Journal of Business and Technical Communication*, 7(1), 12–36. <https://doi.org/10.1177/1050651993007001002>
- St, & Germaine-Madison, N. (2006). Instructions, visuals, and the English-speaking bias of technical communication. *Technical Communication*, 53, 182–190.
- Takaki, R. (2008). *A different mirror: A history of multicultural America*. Back Bay books/Little, Brown and Company.
- Tufte, E. (1990). *Envisioning information*. Graphics Press.
- Urla, J. (1993). Cultural politics in an age of statistics: Numbers, nations, and the making of basque identity. *American Ethnologist*, 20(4), 818–818. <https://doi.org/10.1525/ae.1993.20.4.02a00080>
- Walker, F. A. (Comp.). (1874). *Statistical atlas of the United States based on the results of the ninth census 1870*. U.S. Census Office. Julius Bien.
- Walker, F. A. (2006). Immigration should be halted. In L. K. Egendorf (Ed.), *Immigration* (pp. 81–95). Thomson Gale. (Reprinted from Restriction of Immigration. Atlantic Monthly, by F.A. Walker, June 1896).
- Ware, C. (2013). *Information visualization: perception for design* (3rd ed). Elsevier.
- Welhausen, C. A. (2015). Power and authority in disease maps: Visualizing medical cartography through yellow fever mapping. *Journal of Business and Technical Communication*, 29(3), 257–283. <https://doi.org/10.1177/1050651915573942>.
- Williams, M. F. (2010). *From black codes to recodification: Removing the veil from regulatory writing*. Baywood.
- Williams, R. (2015). *The non-designer's design book: Design and typographic principles for the visual novice*. Pearson Education.
- Williams, M. F., & Pimentel, O. (Eds.). (2014). *Communicating race, ethnicity, and identity in technical communication*. Baywood Publishing Company.
- Wolfe, J. (2015). Teaching students to focus on the data in data visualization. *Journal of Business and Technical Communication*, 29(3), 344–359. <https://doi.org/10.1177/1050651915573944>.

Zhang, Y. (2017). Assessing attitudes toward content and design in alibaba's dry goods business infographics. *Journal of Business and Technical Communication*, 31(1), 30–62. <https://doi.org/10.1177/1050651916667530>.

Author Biography

Li Li is an assistant professor of Professional Writing and Rhetoric in the Department of English, Elon University. Her research focuses on data visualization, multimedia and visual rhetorics, and international rhetorics.